



Implicit theories of leadership: Stability and change over two decades

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ABSTRACT

Much has changed in the last 20 years, but have people's naïve conceptions of leaders changed as well? Paralleling Offermann et al.'s (1994) study of the content of implicit leadership theories with new samples, the present study investigates ILT stability and change across a 20-year period. Results indicate that, as in 1994, Sensitivity, Dedication, Tyranny, Charisma, Strength, Masculinity, and Intelligence were confirmed as ILT factors. Analyses revealed a new factor, Creativity, and the rearranging of some characteristics across factors. The nine-factor, 46-item scale was confirmed with an independent sample, yielding superior fit indices to the eight-factor solution. This supports the view of ILTs as having both remarkably stable elements despite organizational and societal changes as well as contextually-sensitive elements. Open-ended characteristics had no references to females despite reference to males, as in 1994; thus, “think leader, think male” appears to persist in terms of naïve conceptions of leadership.

Introduction

Interest in how people perceive and respond to leaders continues to remain strong. The implicit and naïve conceptualizations people hold of leaders – their implicit leadership theories (ILTs) – represent the cognitive structures or schemas that specify what people expect from leaders in terms of leader traits or attributes. Perhaps most importantly, ILTs have been shown to relate to perceptions of actual leaders (e.g., Schyns, Felfe, & Blank, 2007; Sy et al., 2010). For example, leaders exhibiting prototypic ILTs have been seen as more transformational in their leadership style (Bass & Avolio, 1989). Yet ILTs can bias leadership ratings (Gioia & Sims, 1985), distorting memory and resulting in perceivers inaccurately reporting category-consistent leader behaviors that did not actually occur (Shondrick & Lord, 2010). Individual judgments of archetypal leadership traits have been shown to influence our standards for leaders, including behavioral expectations (Lord, Foti, & De Vader, 1984). More recently, differences between follower ILTs and the recognition of these attributes in a leader have been associated with poorer quality relations with the leader (Topakas, Martin, & Epitropaki, 2015). In addition, congruence between leader and follower ILTs can influence both parties' assessment of the quality of their relationship (e.g., Riggs & Porter, 2017; Tsai et al., 2017). Thus, people's naïve views of leaders may set the course for their interactions with the leaders with whom they come into contact in daily life.

ILTs stem conceptually from leader categorization theory, which postulates that people form mental schemas of leaders based on perception (Lord et al., 1984; Lord, Foti, & Phillips, 1982). Individuals use

leader schemas to process information and identify leaders through three hierarchical levels: superordinate, basic, and subordinate (Rosch, 1978). At the highest, most general, *superordinate* level is the simple classification of a stimulus person as either leader or non-leader. The *basic* level is often considered to be most important in that it typically reflects the names most associated with objects (Mervis & Rosch, 1981). In terms of leadership, basic level categories consider the context of leadership; 11 examples such as business leader, political leader, and military leader categories have been identified in previous research (Lord et al., 1984). Traits attributed to leaders typically varied with leadership categorization and context, so that of 59 possible attributes only intelligence was found to be seen as a critical feature across most (10 of 11) basic categorizations (Lord et al., 1984). The lowest level of categorization, *subordinate*, is the most specific, where a military leader might be categorized as an Army major or a Navy admiral.

A distinction exists between recognition-based and inference-based leader processes. Recognition-based processes emphasize leader identification through categorization, as described above. Under inference-based processes, people distinguish leaders from others based on the leader's behavior, rather than perceived traits. Lord and Maher offer their own definition of leadership as “the process of being perceived by others as a leader” (Lord & Maher, 1991, p. 11). According to this definition, the mere perception of leadership, whether based on behaviors or traits, is enough to influence others. Inference-based processes focus on the outcomes of leadership (Lord & Maher, 1991), suggesting that group performance, especially group success, is intertwined with people's inherent definitions of leaders. A 1982 content analysis of news

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sources found evidence of differing perceptions of “leaders” and “effective leaders” (Lord et al., 1982). Positive items were shown to be viewed as more prototypic of “effective political leaders” as opposed to “political leaders” in general (Foti, Fraser, & Lord, 1982). Offermann, Kennedy, and Wirtz (1994) examined perceptions of leaders, effective leaders, and supervisors, and found that ratings for leaders and effective leaders were typically similar and more positive than ratings of supervisors; however, effective leaders were viewed as less tyrannical than either leaders or supervisors.

Twenty years ago, based on this work on leader categorization, Offermann et al. (1994) investigated the content and structure underlying ILTs through a rigorous, multi-phase process using samples of students and working adults. They identified eight specific ILT factors: Sensitivity, Dedication, Tyranny, Charisma, Attractiveness, Masculinity, Intelligence, and Strength. This frequently cited work resulted in a 41-item ILT scale that has been used as the basis for other subsequent investigations into ILTs (e.g., Epitropaki & Martin, 2004; Epitropaki & Martin, 2005; Keller, 2000).

Epitropaki and Martin (2004) noted that mental representations like ILTs may be dynamic and subject to change when the context changes, arguing for the importance of longitudinal assessment. Thus, they investigated the generalizability of Offermann et al.'s work, with the hopes of also reducing the number of scale items. On the basis of their results, they proposed a six-factor structure comprised of Sensitivity, Intelligence, Dedication, Dynamism, Tyranny and Masculinity, dropping the Attractiveness factor as prototypic but not “core,” and collapsing the Strength and Charisma factors into a Dynamism factor. Their work provides good evidence for structural stability of ILTs 10 years after the original Offermann et al. (1994) study, and also provided support for the stability of ILTs over a 12-month period. It also supports the generalizability of ILTs across different groups of workers in terms of age and organizational positions.

More recent theoretical work on ILTs has argued for the more complex and dynamic nature of leadership prototypes, noting that they can be both sensitive to context and yet still produce stability over time (e.g., Lord, Brown, & Harvey, 2001). This suggests that some aspects of ILT content as described by Offermann et al. (1994) and Epitropaki and Martin (2004) may have remained stable over time, while some aspects may have been discarded and new categories added. The purpose of the present study was to investigate the stability and possible changes in ILTs across a 20-year period by following the procedures used in Offermann et al.'s (1994) original study of the content of implicit leadership theories using new samples while allowing for the addition of new attributes and/or the removal or realignment of old ones.

The prospect of both stability and change in ILTs is supported by recent theorizing about ILTs. Advancing a connectionist perspective that proposes a two-way exchange between leaders and followers that can be adjusted to fit changing contexts, and which shows variability within and across individuals, Lord et al. (2001) suggest that this model “can be used to understand both the stability and flexibility that is witnessed in the application of leadership prototypes” (p. 311). Once formed, leadership perceptions are believed to be slow to change, but can be modified through a process of unsuccessful matches of stimulus and prototype that may proceed at different rates for different perceivers. Using adaptive resonance theory (ART, Grossberg, 1999), Shondrick and Lord (2010) argue that there are individual differences in the strictness of individual profile matching, with some perceivers demanding a tighter match and others more willing to accept a looser fit in order for someone to be determined to fit the leader prototype. As new experiences with leaders are encountered, the contextual meaning attached to a particular leader may change without initially changing views of leaders in general, making ILTs less likely to change quickly. However, a number of unsuccessful leader-prototype matches encountered over time may impact views of leaders in general. Thus, ILTs can be both stable and mutable, with new schema being created when matches to existing schema cannot be found while matched schema

continue to persist over time.

This connectionist perspective has also been used to examine implicit theories of followership (IFTs, Braun, Stegmann, Bark, Junker, & van Dick, 2017; Shondrick & Lord, 2010; Sy, 2010), which are the beliefs that people hold about the prototypical characteristics of followers. More recently, Lee, Martin, Thomas, Guillaume, and Maio (2015) suggested viewing leadership perceptions as attitudes towards leaders and that future work is needed to consider implicit attitudes towards leadership and how they impact follower outcomes. As leadership research increasingly views leadership as a socially-constructed reciprocal exchange between leaders and followers, understanding perceptions of both types of actors becomes critically important.

Although leadership perceptions are typically backward-looking, based on past leader behaviors and outcomes, perceivers typically see such perceptions as reflecting stable leader characteristics that are expected to generalize to the future (Lord & Dinh, 2014). Thus, examining the content of ILTs is still important in understanding the frame in which followers begin engaging with leaders and which sets expectations for subsequent interactions where individuals will attempt to match leader stimuli to the expected prototype.

The changing workplace

In keeping with previous theorizing, we argue that major changes in the environmental demands on leaders may change people's perceptions of the characteristics necessary for successful leadership over time, adding new elements and discarding others deemed no longer relevant. In the organizational context, this means that some aspects of ILTs may change along with aspects of a changing workplace while other aspects remain stable. In considering the potential for organizational context to change prototypic expectations, it is clear that the context of the modern workplace has changed markedly in the > 20 years since Offermann et al.'s (1994) original work, and especially in the 13 years since Epitropaki and Martin's (2004) study. Many of these changes have occurred broadly across work sectors and, indeed, across national boundaries, affecting a large majority of working adults. These workplace changes may well have affected the ways in which people view leaders.

One major shift within the U.S. workforce has been the increasing presence of women in organizational management roles. Now that women occupy about half of U.S. management jobs (Catalyst, 2014), it is possible that the traditional “think manager, think male” mantra noted years ago by Virginia Schein (1973; Schein, Müller, Lituchy, & Liu, 1996) now often described as “think leader, think male” - may be eroding in naïve conceptions of leadership. A detailed meta-analysis by Koenig et al. in 2011 concluded that although the overall masculinity of leader stereotypes continues, this masculine construal of leadership has decreased over time and is greater for male respondents than for females. As a result, the factor of Masculinity, found by both Offermann et al. (1994) and Epitropaki and Martin (2004) is one where some change in this specific ILT may be likely while other factors remain stable. In the Offermann et al. (1994) study, open-ended leader characteristics generated by participants, over half of whom were women, had no references to the female gender despite references to men and masculinity. Epitropaki and Martin (2004) similarly found masculinity represented in ILTs. However, as more people currently work for and with both women and men in leadership roles, the association of leadership solely with one gender may have decreased.

Numerous authors have suggested that women may, in fact, be better leaders than men (i.e., Helgesen, 1995; Wilson, 2004), rating women significantly higher than men on desirable transformational leadership characteristics such as idealized influence, inspirational motivation, individualized consideration, and contingent reward as well as lower than men on the less desirable management-by-exception and laissez-faire styles (Eagly, Johannsen-Schmidt, & van Engen, 2003). As enthusiasm for the use of teams in organizations continues to rise

(Mathieu, Maynard, Rapp, & Gilson, 2008), leadership characteristics may now have come to be associated with some traditionally female attributes such as greater attraction towards cooperation and teamwork (e.g., Kuhn & Villeval, 2013) as well as some traditionally male ones (such as tough). It is also possible that the salience of gender as a leadership trait has weakened to the point of its removal from ILTs entirely. However, given that gender equity at senior organizational leadership levels has yet to be achieved, with men still dominant in senior leadership roles (Catalyst, 2015), we suspect that gender is still in people's minds when they think of leaders, but that perhaps ILT content has become more androgynous, containing elements of both traditionally male and female characteristics.

Hypothesis 1. The increasing gender diversity in leadership ranks in the U.S. will be reflected in the representation of both traditionally male and traditionally female-associated characteristics in ILT content.

In addition to the possible addition of feminine aspects of leadership, there may be additional ILT components based on broadly experienced workforce changes that place new demands and expectations on today's leaders. One major change over the last ten years has been the tremendous impact of technology on our lives, both at home and at work, with employees engaging heavily with advanced technologies that can change the nature of workplace interactions, including interactions between leaders and followers. For example, Facebook was created in 2004, the year of Epitropaki and Martin's replication of the original Offermann et al. study, and has now grown to include over 1700 million subscribers. As of June 2017, 52% of the world's population had internet access (Internet World Stats, 2017). Many more employees telework all or part of the time, with a recent survey reporting that 43% of Americans spend time working remotely (Chokshi, 2017), forcing more leader-follower interactions to be virtual rather than face-to-face. This change in communication mode may challenge conceptions of leaders and their roles and has given rise to discussions of what "e-leadership" entails and how it may differ from traditional face-to-face leadership (Avolio, Sosik, Kahai, & Baker, 2014). Whether such changes in leader demands in our technological world affect ILT content has yet to be determined.

Further, the growing gender diversity in the ranks of organizational leaders is also coupled with the presence of an increasingly diverse workforce that may demand new leadership skills. Leaders are now charged with developing and encouraging staff from a variety of backgrounds, giving rise to calls for "inclusive leadership" (Gallegos, 2014; Hollander, 2009; Sugiyama, Cavanaugh, van Esch, Bilimoria, & Brown, 2014) and intercultural competence (Bennett, 2014; Leung, Ang, & Tan, 2014) as core leadership skills. Yet even sensitivity to diversity may result in additional issues, where leaders are reluctant to give feedback to those different from them for fear of seeming unsupportive, depriving those individuals of opportunities to improve (Offermann & Basford, 2014). Whether awareness of the need for leaders to be more inclusive of diversity and to be able to lead even those markedly different from oneself has permeated the schema of an effective leader remains to be seen.

These changes in technological advancement, and leader and follower diversity, suggest possible areas in which leadership contexts may have changed sufficiently to have infused naïve leadership conceptions with new factors associated with ideal leaders that are not represented in previous ILT factors. Thus, we predict that:

Hypothesis 2. Changes in workplace technology may be reflected in changes in ILT content, whereby leaders are expected to be facile in such environments.

Hypothesis 3. Greater attention to diversity issues may be reflected in a concern about more inclusive leadership.

Increasing globalization represents an additional shift in the modern workplace that might affect current perceptions of ideal leaders. There

is evidence to suggest that leader prototypes vary by culture, prompting attention to cultural differences in leadership perceptions (Gerstner & Day, 1994). The work of the well-known GLOBE study (House, Hanges, Javidan, Dorfman, & Gupta, 2004) comparing views of leadership in 58 countries shows that at least some dimensions, such as Charisma, appear to be widely endorsed across cultures. Other work has shown that the content of implicit leadership theories differs among Chinese and American respondents. Identification of separate Chinese ILTs suggests that Western leadership theory may not generalize to more collectivist cultures (Ling, Chai, & Fang, 2000). An additional result from the same study also found discrepancies in ILTs among three Chinese social groups, further highlighting the fact that ILTs can differ between social and cultural groups, even within the same country. As a result, research on ILTs needs to be conducted within single country contexts, with attention to within country social subgroups as appropriate, as well as through cross-country comparisons. For the present study, we focus on ILTs as experienced within the U.S., as was done in Offermann et al.'s (1994) study.

As noted earlier, once formed, leadership perceptions are believed to be slow to change, suggesting that many ILT components are likely to remain stable for long periods of time. Apart from possible additions due to changing workplaces, and changes in the Masculinity factor described above, the other seven of Offermann et al.'s (1994) original eight factors appear to be equally relevant to the modern workplace as they were twenty years ago, representing core elements of perceived leadership that are likely to be stable and resistant to change. Sensitivity, Dedication, Tyranny, Charisma, Attractiveness, Intelligence, and Strength all are attributes regularly discussed in current treatises on effective leadership, as in previous years. We have seen no suggestion that modern perceivers would not continue to find them to be prototypic, although the specific items characterizing these attributes may have changed with current language. Thus, given the previous degree of stability found from 1994 to 2004, and the lack of specific reasons for a change in perceived prototypicality, we hypothesize that the original eight factors, including Masculinity, will continue to be displayed in current samples, with the prospect of additional content reflecting feminine aspects of leadership and/or technological and diversity concerns added to these core, stable ILTs.

Hypothesis 4. The original eight ILT content domains from 1994 will continue to be represented in participant ILTs twenty years later.

General method

In order to assess change in individual's implicit theories of leadership over the last 20 years, we paralleled the procedures used by Offermann et al. (1994), using four phases of data collection with different samples. In the first phase, a sample of undergraduate students individually generated a list of items believed to be characteristic of a leader, as was done in the 1994 study. This list was reduced to generate a 170-item questionnaire to be used in subsequent phases. Unlike in 1994, the present study also replicated this process with an adult working sample to insure generalizability beyond college samples. The second phase used an adult working sample to confirm the 1994 structure that underlies ILTs with the same 41 items used in 1994; a third phase explored whether a better fit was possible using the full 170-item 2014 data. Following the development of an augmented structure based on both 1994 and 2014 data, the fourth phase consisted of collecting a second student sample to confirm the resulting factor structure.

Phase 1: item generation

Method

Sample and procedure. The sample for generating the items consisted of 121 undergraduate students. The sample was 31.4% male and 68.6%

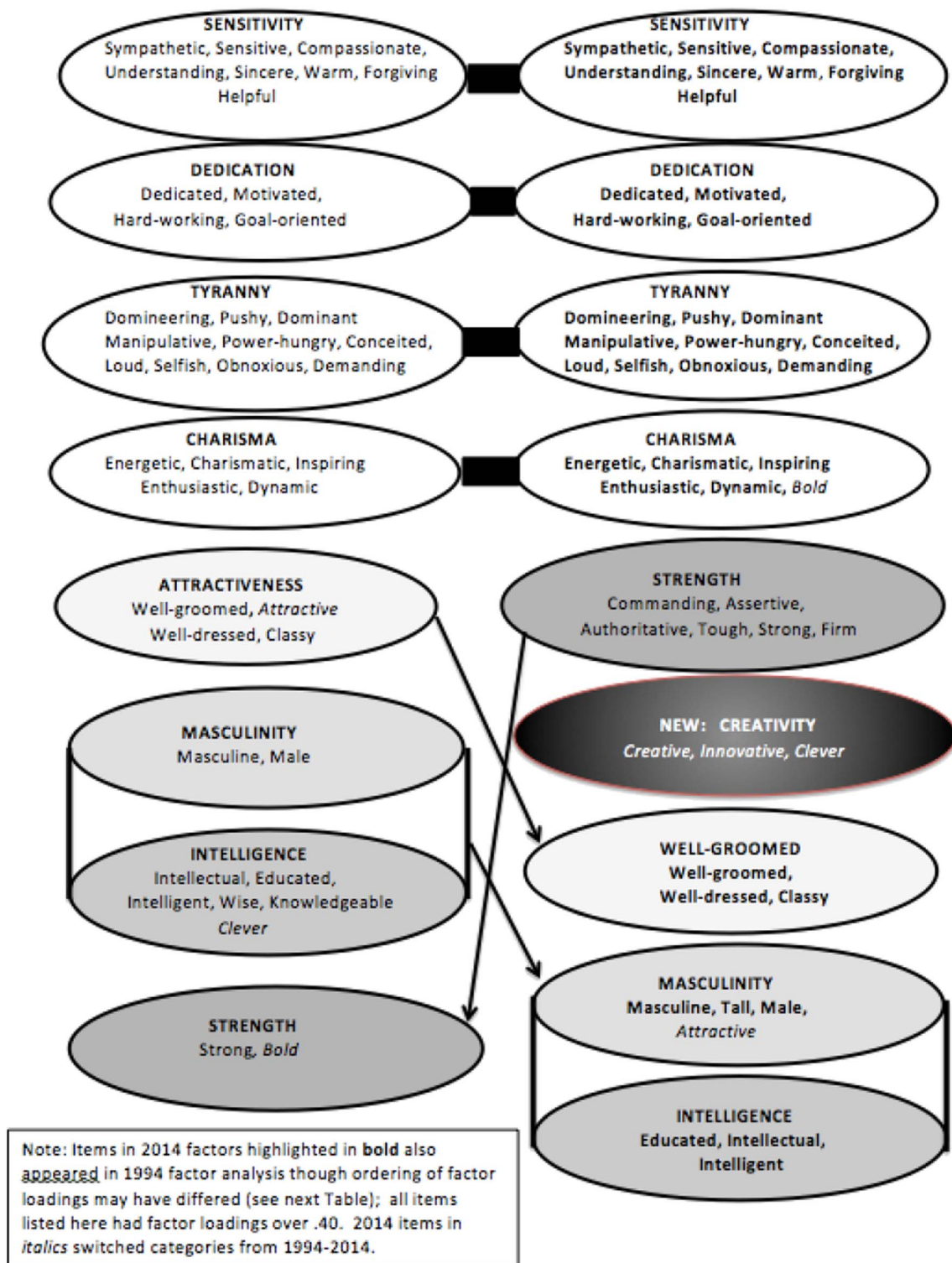


Fig. 1. ILT factors from 1994 (left) and from 2014 (right).

female. Participants were given a sheet of paper with 25 blank lines and were asked to list up to that many traits or characteristics that they considered to be characteristic of a leader. The students were not given a definition of the word leader.

Results

This sample resulted in a list of 426 different words or phrases that were generated by one or more students. To reduce the number of items for the second phase of testing, the frequency that each item was

generated was computed for each of the words and items that were generated only once or twice were eliminated. Only one of redundant items considered fully synonymous in meaning was retained. For example, “adaptive” and “adaptable” were not both retained; adaptive was chosen for retention because it was the term generated more often. This process resulted in a total of 170 items to be used for the validation and factor identification. There was considerable overlap between this list of 170 items and the 160 items found in the 1994 analyses, with all 41 of the final set of items from the 1994 study included among the

Table 1
Model fit statistics.

Measure	χ^2	Df	p	RMSEA (95% CI)	CFI	SRMR
Original CFA model	3328.560	8751	< 0.001	0.069 (0.066; 0.071)	0.852	0.094
Epitropaki and Martin (2004) CFA model	1937.060	411	< 0.001	0.07 (0.067; 0.074)	0.881	0.074
New CFA model	2308.746	953	< 0.001	0.056 (0.053; 0.059)	0.866	0.063

170. In total, 76% of the items from the original 1994 sample were present in the current sample. For some of the original items that were not present, a very similar item was generated. For example, the original sample contained the item “able to handle pressure” while the present sample contained items such as “calm under pressure.” However, there were also some original items that dropped out, and some new additions. New item additions included “inclusive,” “culturally competent,” “emotionally intelligent,” “compassionate,” “passionate,” “team builder,” “empowering,” and “relatable,” terms that are clearly more common in current vocabulary than in previous years. Items from 1994 addressing both ethics and creativity surfaced again, and appeared in the item list, with the new addition of “visionary” to the possible creativity domain.

Although our present work parallels the original Offermann et al. methodology using an initial student sample as they did, we also collected data from an additional 125 working adults using Amazon's Mechanical Turk (MTurk), a popular crowdsourcing tool now widely used in research. These online contract labor portals have been shown to provide data that are as or more reliable than university student samples while providing access to older, more diverse adults with more work experience (Behrend, Sharek, Meade, & Wiebe, 2011). We used recommended MTurk data cleaning procedures (DeSimone, Harms, & DeSimone, 2015), including removing those who spent insufficient time on each item and duplicate responses. On this basis, one participant (0.08%) was removed, resulting in a final sample of 124. A team of two research assistants received training on identifying content areas and duplicate items. After there was sufficient agreement in training sessions, the coders went through both lists of items and determined that there were no additional content areas generated from the adult sample that were not present in the student sample. Thus, it does not seem to matter whether Phase 1 uses students or working adults in the generation of initial items for testing. The resulting 170 item scale was then used in subsequent study phases described below.

Phase 2: original factor confirmation

Method

Sample and procedure. Participants were 860 adults, 40% male and 60% female, between 18 and 76 years of age with an average age of 37.5 years. Most of the sample (73.4%) was employed full or part time and 17.4% were full or part time students. Participants were 76% White/Caucasian, 6.6% Asian, 8.7% African American/African, 5.1% Hispanic, 0.9% Native American, and 2.1% other. The sample was recruited using Amazon's MTurk and was given a survey of the items that were generated in Phase 1. Instructions asked participants to indicate the extent to which each of the 170 descriptors generated in the previous phase were characteristic of a leader from “not at all characteristic of a leader” to “extremely characteristic of a leader.” Each item was rated on a 10-point scale to mirror the metric used in the original 1994 study. The same data cleaning procedures described above were used to eliminate participants that failed attention checks or did not spend enough time completing the survey. This process

yielded 860 participants for analyses.

Results

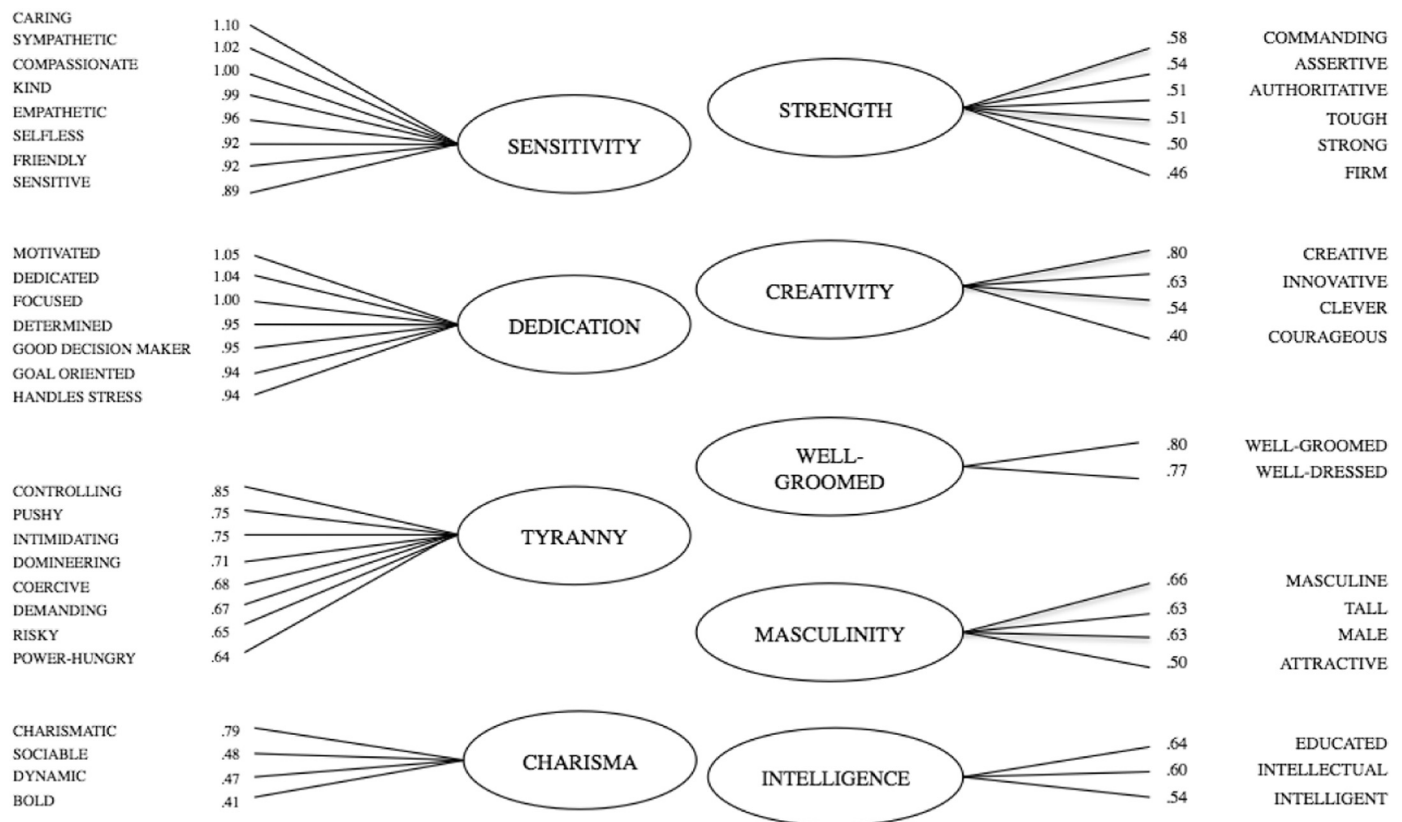
The first step was to see if the new adult sample confirmed the original factor structure from 1994 to see if the structure remained stable. The final 1994 factor structure had eight factors composed of 41 items: Sensitivity with eight indicators, Dedication with four indicators, Tyranny with ten indicators, Charisma with five indicators, Attractiveness with four indicators, Masculinity with two indicators, Intelligence with six indicators, and Strength with two indicators. Fig. 1 displays the final factor structure from 1994.

For the initial confirmatory analysis, we extracted the same 41 items from the new set of 170 to see if those items alone would confirm the 1994 structure. Results from this analysis yielded a moderate level of fit and can be seen in Table 1. The data were very close to fit recommendations set forth by Hu and Bentler (1998, 1999) and deemed acceptable. This supports Hypothesis 4 in that the 2014 structure supports the retention of the original 1994 ILT factors, but also suggests potential room for improvement by incorporating additional items or factors.

Post hoc analyses. While the Offermann, Wirtz, and Kennedy model is the most cited example of ILT structure, the 2004 examination by Epitropaki and Martin revision of the original structure has also received considerable attention. In order to be thorough, this revised factor structure was also tested. Results from the CFA on the Epitropaki and Martin (2004) revised scale also had acceptable fit which is reported in Table 1. Some of the fit indices showed a slight improvement in fit over the Offermann et al. model, likely due to the simpler structure. Because these models are not nested, a direct statistical comparison between the models could not be made. However, because this model also failed to receive ideal fit, it still suggests that there could be some changes to the model in the time since the revision was published.

Phase 3: new factor identification

Although the factor structure largely confirmed what was found in 1994, we sought to determine if any new factors had emerged using the 2014 item set and whether that might improve model fit. The 2014 item set included the final 41 from 1994, some new items not generated in 1994, and many items that appeared in both 1994 and 2014 item sets. The adult MTurk sample described above was then used to see if the factor structure would change if considered from an exploratory perspective rather than a confirmatory perspective. An exploratory factor analysis with principal axis factoring and promax rotation allowing for intercorrelated factors was conducted. There were 24 factors with an eigenvalue over one; investigation of the scree plot suggested a cut off of between 8 and 9 factors. Based on the interpretability of the factors we retained the nine factors depicted in Fig. 1, shown in comparison with their 1994 counterparts. The first factor, Sensitivity, comprised 36.6% of the total variance; the following eight factors cumulatively comprised 59% of the total variance. Fig. 2 shows the items with highest loadings on each of the nine factors. Although this structure was very similar to the original 1994 structure, there were also some key differences, notably the emergence of a new factor, Creativity. This supports the general contention that the ILT structure may now include additional elements, in this case the elevation of Creativity into a new factor, but does not address changes in leader-member interactions as mediated by technology. Because female/feminine items were not naively generated, Hypothesis 1, suggesting the incorporation of female and feminine into ILTs could not be supported.



Note: Numbers adjacent to manifest variable names represent the highest factor loading of the item.

Fig. 2. Factor structure of implicit leadership theory model derived from 2014 sample.

Phase 4: new factor confirmation

Having now generated a new nine-factor solution using 2014 data, we sought to confirm this structure with an additional sample.

Sample and procedure

This sample consisted of 538 undergraduate students in various psychology classes. The sample was 31% male and 69% female with an average age of 19.4 years. Participants were 67.6% White/Caucasian, 17.7% Asian, 7% African/African American, 3% Hispanic, 0.6% Native American, and 4.1% other. Participants were again given a survey that contained all of the generated items from the previous phase. Instructions asked that participants rate each descriptor item on a 10-point scale from “not at all characteristic of a leader” to “extremely characteristic of a leader.” Participants rated all items and those ratings were then used for the final confirmatory factor analysis of the newly discovered factor structure from the previous phase of data collection.

Results

Paralleling the 1994 Offermann et al. process, we reduced the 170 items by choosing the items with the highest loadings from each factor in order to best represent each factor for the confirmatory factor analysis (CFA). This resulted in eight indicators for Sensitivity, seven indicators for Dedication, eight indicators for Tyranny, four indicators for Charisma, six indicators for Strength, four indicators for Creativity, two indicators for Appearance, four indicators for Masculinity, and three indicators for Intelligence. Listwise deletion was used to handle missing data.

Model fit for the confirmatory factor analyses was again done using recommendations put forth by Hu and Bentler (1998, 1999). Per their

recommendations, the fit of the CFA was determined using the Root Mean Square Error of Approximation (RMSEA), the Comparative Fit Index (CFI), and the Standardized Root Mean Square (SRMR), parsimonious, incremental fit, and absolute fit, respectively. Given our large sample size, it was not surprising that the chi-square statistic was found to be significant, as this value is affected by sample size. Although comparative fit indices penalize for model complexity, the obtained value for CFI was only slightly outside of the recommended criteria, a level of fit commonly deemed “acceptable” (Ambrose & Schminke, 2003, p. 299; Lent, Lopez, Lopez, & Sheu, 2008, p. 57) and “adequate” (Karim & Behrend, 2013). The RMSEA and SRMR values both fell within Hu and Bentler’s recommendations. The values obtained from the CFA can be seen in Table 1. Compared to the original 8-factor 41-item solution, this 9 factor, 46-item solution improved model fit and was retained as the final ILT scale.

Post hoc analyses

As in 1994, spontaneously generated leader characteristics included terms associated with men but not of women, specifically “male” and “masculine.” Although not included in the factor analyses or CFAs since they were not naively generated, we included the characteristics “female” and “feminine” as part of the 170 item scale to see how those terms related to the emerging factors in comparison to their male counterpart items. Interestingly, we found that in the adult sample “female” and “feminine” correlated most highly with endorsement of the “male” and “masculine” items. This may be an indication of some movement towards a less specific gender association for leaders despite the fact that “female” was not implicitly generated the way “male” was. Endorsement of all four gender items (i.e., female, feminine, male, and masculine) were significantly and positively associated with ratings of

Table 2
Factor correlations with adult sample.

Measure	Mean	SD	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Female	5.07	2.54	-												
2. Feminine	4.27	2.34	0.59**	-											
3. Male	5.29	2.77	0.40**	0.39**	-										
4. Masculine	5.20	2.86	0.20**	0.21**	0.56**	-									
5. Sensitivity	7.65	1.67	0.09	0.02	-0.12	-0.11	(0.92)								
6. Dedication	8.90	1.28	-0.14*	-0.20**	-0.08	-0.07	0.54**	(0.92)							
7. Tyranny	5.32	1.87	0.26**	0.18*	0.35**	0.45**	-0.16*	-0.11	(0.88)						
8. Charisma	8.24	1.45	0.03	-0.05	0.00	0.09	0.50**	0.64**	0.09	(0.78)					
9. Strength	8.02	1.52	0.04	-0.09	0.14	0.17*	0.32**	0.60**	0.38**	0.62**	(0.84)				
10. Creativity	8.08	1.39	0.06	-0.05	-0.04	0.01	0.70**	0.65**	-0.03	0.64**	0.52**	(0.82)			
11. Well-groomed	7.00	2.24	0.20**	0.13	0.22**	0.13	0.34**	0.26**	0.25**	0.39**	0.37**	0.36**	(0.79)		
12. Masculinity	5.06	2.12	0.42**	0.41**	0.80**	0.79**	-0.13	-0.12	0.49**	0.11	0.16*	-0.03	0.29**	(0.82)	
13. Intelligence	8.43	1.44	-0.01	-0.07	-0.09	0.03	0.55**	0.61**	-0.04	0.56**	0.48**	0.69**	0.34**	-0.05	(0.80)

Note. N = 860. Reliabilities are listed in bold, italicized parentheses on the diagonal.

** $p < 0.01$, two tailed.

* $p < 0.05$, two tailed.

Tyranny, though the associations were larger with the male and masculine items. The items “female” and “male” were also significantly positively correlated with Well-Groomed. Results are shown in Table 2, along with means and SDs for all nine factors.

As with Epitropaki and Martin (2004), additional post hoc analyses to determine if factor invariance existed across the sex of participants was not deemed appropriate due to limited subgroup sample size and highly inequivalent sample size between groups (Barrett, 2007; Meade & Bauer, 2007; Meade, Johnson, & Braddy, 2008; Schmitt & Kuljanin, 2008). However, to explore the prospect of differing factor structures by gender, we conducted separate exploratory factor analyses with the larger group of male and female participants from Phase 3. Comparison of these data revealed some interesting patterns. All nine factors were represented in both male and female samples, with the items chosen to represent the factors shown in Fig. 2 generally aligning with the appropriate factors, suggesting broad agreement across gender on most factors. The major exception was for Charisma and Strength, which formed a single factor for women, more similar to Epitropaki and Martin's (2004) dynamism factor. For men, Strength was its own factor, with items from the Charisma factor (“charismatic” and “sociable”) loading with the items good public speaking, outgoing, and vocal rather than bold and dynamic – more of an effective communicator factor. The Creativity factor appeared as the fourth extracted factor with very high loadings for men; Women had creativity items loading in the 0.33–0.45 range aligned with the Charisma/Strength factor. Men had “tall” and “attractive” as a separate factor, whereas women loaded those items with “male” and “masculine.”

Another interesting finding was that although both men and women had Sensitivity and Dedication as their first two factors, with all the items as indicated in Fig. 2 loading on them for both genders, they were extracted in different orders. Sensitivity was the first and strongest factor for women, followed by Dedication; for men, Dedication was the first and strongest factor, followed by Sensitivity.

As in 1994, the ethical items did not cohere into a significant separate factor in ILTs, either with the full sample or examining the gender subsamples. Looking back on the initial stage of item generation, ethical items emerged, but were rarely among the first items generated on people's lists. Rather, these items loaded on the Sensitivity factor (for both genders), with loadings far lower than those used to comprise that factor, which were above 0.8. Although lower, several ethics-related items loaded above the 0.4 cut-off often used to be included in a factor (fair 0.58, honest 0.54, ethical 0.44). Thus, ethics seems to be incorporated into people's views of a sensitive, considerate leader rather than being seen as a separate descriptor.

Discussion

Twenty years after the initial Offermann et al. (1994) study investigating what comprises individual's implicit theories of leadership it appears that the overall structure of what people consider to be characteristic of leaders remains largely unchanged. In light of charges against psychological research as lacking in replicability (Open Science Collaboration, 2015), these data, along with that presented by Epitropaki and Martin (2004), provide compelling evidence of the continuing pattern of naïve conceptions of leadership as held by the general public. As in the original study, factors of Sensitivity, Dedication, Tyranny, Charisma, Strength, Masculinity, and Intelligence emerged in confirmatory analyses using the same 41 items as in the original study with a new sample of working adults. However, conducting an exploratory factor analysis revealed that there are also a few key differences from the factors that emerged in the original study, including the emergence of a new ninth factor, Creativity, and the subtle rearranging of some characteristics within the factors (see Fig. 2). A comparison of factors from the original study and the current study can be seen in Fig. 1. This nine-factor, 46-item scale was then confirmed with an independent sample and showed fit indices that were superior to those of the eight-factor solution.

The most noticeable difference between 1994 and 2014 solutions is the emergence of a Creativity factor. This factor consisted of traits such as creative, innovative, and clever – all items that were cited by some individuals in the original item generation in 1994 but which did not coalesce into a coherent factor at that time. Post-hoc exploratory analyses showed this to be a stronger element for men compared to women. Finding evidence of both the stability of most factors and the addition of a new factor supports the view of ILTs as having both stable and contextually-sensitive elements (Lord et al., 2001).

There are several possible reasons for the emergence of this particular new factor. Although older research placed “creative” on lists of non-leader attributes (Lord et al., 1984), as organizations now place an increased emphasis on innovation as key to future success, creativity may presently be seen as a more important aspect for effective leadership than in previous years. In fact, creativity was recently noted as a striking absence as a content area in ILTs by Epitropaki, Sy, Martin, Tram-Quon, and Topakas (2013). The elevation of creativity found here may also be evidence of a possible “tech effect,” predicted in Hypothesis 3. Although the impact of technology on organizational life was not directly seen in a new ILT factor, the increasing importance of technology both in the workplace and almost every aspect of people's lives (Thomas, 2014) may be increasing beliefs that creativity and innovation is a required component of successful leadership. The media has highlighted (even glorified) high-tech leaders such as Steve Jobs,

attributing his success and that of his company to his creativity, imagination, and entrepreneurial abilities (Cornelissen, 2013). Extensive media exposure to Jobs and other visionary leaders may be one context factor driving creativity into greater prominence when people think about leaders. Indeed, “visionary” was a new addition to the 2014 item set that was not generated in 1994. The combination of items reflecting creativity into a separate factor may point to an increasing belief in the importance of innovation to leadership roles, perhaps influenced by the modern adulation of technology.

The increasing acknowledgment of the importance of creativity, particularly by men, may not be good news for women leaders. A recent series of five studies by Proudfoot, Kay, and Koval (2015) using both experimental and archival data showed that so-called “outside the box” creativity was more strongly associated with traditionally masculine characteristics than with stereotypically feminine characteristics, that men’s ideas are evaluated as more ingenious than women’s, that identical output is seen as more creative when attributed to a man than to a woman, and that supervisors evaluate female executives as less innovative than their male counterparts. These differential assessments of creative potential based on gender may be particularly limiting to women in male-dominated occupations, and potentially explain their limited numbers in STEM fields and the burgeoning technology sector.

As in 1994, the ethical items did not cohere into a significant separate factor in ILTs. Ethical items emerged during the initial stage of item generation, but were rarely among the first items generated on people’s lists. Although it could be argued that ethical behavior is represented as the obverse of the Tyranny factor – that perhaps ethics and integrity are considered only when its values are not upheld by leaders who use their power to oppress and dominate – our data does not support that explanation. Examination of correlations between potential ethical leadership items (ethical, integrity, honest, and trustworthy) shows correlations between these items and the Tyranny scale ranging from -0.03 to -0.14 . Rather, ethical items loaded with the Sensitivity factor, suggesting that leaders viewed as sympathetic, understanding, and caring are also likely to be seen as those with greater integrity.

There was no mention in the items of the need for leaders to have special “e-leadership” skills. Leadership in its public perception is apparently still viewed as people-centered. Although leaders doubtless make greater use of email and other technologies to interact with staff members than in previous years, the importance of face-to-face interaction between leaders and followers is still valued. As Chang (2015) noted, “if the point of business were simply to accomplish as many tasks as possible, then yes, an email would probably do. But that’s not what real leadership is about.” Other than perhaps the Well-groomed factor, many of the core characteristics in people’s minds when they think about leaders deal with those characteristics that come to life in personal interaction with others. Is the leader sensitive to my needs? Will I be inspired and encouraged by a dedicated leader, or dominated by an abusive one?

Inclusive leadership also did not emerge as a significant factor on its own, although items reflecting greater attention to diversity were present, notably in the new “culturally competent” and “inclusive” items. Rather, as with ethics, inclusiveness items loaded on the Sensitivity factor, but with lower loadings than items defining that factor. It appears that the understanding and compassionate attributes central to Sensitivity are assumed to extend to everyone within a leader’s span of control without favoritism.

There were also some shifts in the traits underlying the factors that are different from the originally discovered structure. For example, the trait “bold” moved from the Strength factor from 1994 to the Charismatic factor in the present sample. The current analysis shows Strength to now be more about command and assertion, including remaining firm, whereas boldness appears now to be associated more strongly with individuals who are charismatic, which tends to be associated with dynamism and change. Another interesting shift in traits is the shift of the attractiveness item from the original Attractiveness

factor to the Masculinity factor. Originally “attractive” was associated with other physical appearance/grooming traits whereas now it is associated with the traits “masculine,” “tall,” and “male.” While Epitropaki and Martin’s (2004) 21-item scale drops Attractiveness as prototypic but not “core,” our current analyses suggests that this Well-groomed factor accounted for slightly more variance than Intelligence, a factor widely accepted as part of typical ILTs, as noted earlier. In our highly visually-oriented society, it seems that appearance still counts and attractiveness in leaders is associated with men of above-average physical stature, particularly among women.

Despite occupying approximately 50% of U.S. management jobs (Catalyst, 2014), “think leader, think male,” still appears to persist in terms of naïve conceptions of leadership. Open-ended characteristics generated by our modern student sample, over half of whom were women, had no references to the female gender despite references to men, as was true in 1994. The same pattern held for the adult sample. This is despite meta-analytic evidence of women’s equal or greater leadership competence compared to men (Eagly, 2007) and the fact that the factor accounting for the largest variance in leadership perceptions – Sensitivity – is a characteristic traditionally more associated with women. Thus, our work supports Koenig, Eagly, Mitchell, and Risikari’s (2011) contention that there is still a strong tendency for leadership to be viewed as a masculine domain, and that this construal is endorsed more strongly by men than women.

Our post hoc analyses demonstrated that endorsement of ILT factors may differ by gender. Notably, loadings for items in the Masculinity factor were lower for females than for males and the covariances between factors displayed different patterns across gender. Examining factor structures in gendered subsamples showed some interesting gender differences, particularly in views of Charisma. Women conceptualized Charisma in conjunction with Strength – the commanding, charismatic, dynamic, tough leader – whereas men separated these two. For men, charisma was associated with effective public speaking and persona; command, toughness, and strength were something different. So despite its representation in both male and female views of leadership, what “charisma” means to each may be different. It is also interesting to note that the first factor extracted for each gender, accounting for the most variance, was one more typically associated with their gender: Sensitivity for women, Dedication (which includes the more task-oriented items of goal-oriented, focus, good decision maker) for men. These exploratory results suggest the need for further exploration into the differences between male and female implicit conceptualizations of leaders.

Limitations and future research

Our samples were drawn from two sources: undergraduate populations and adult samples composed of MTurk workers, both from within the U.S. While a majority of the MTurk workers were employed full time, they are also internet-connected, and thus may not be representative of other segments of the U.S. workforce. All forms of convenience samples, whether with organizational samples, college students, or MTurk workers, are potentially subject to range restriction which could potentially limit external validity (Landers & Behrend, 2015). By using different types of samples, and confirming findings across them, we sought to increase confidence in the generalizability of our findings. Nonetheless, replication with less educated samples who have limited or no internet access might be instructive.

As a result of using only U.S. samples, our findings may or may not generalize to other cultural contexts. However, despite local differences, work from the GLOBE study (House et al., 2004) shows that there are some leadership dimensions that appear to be widely endorsed across cultures. Some of these dimensions, such as Charisma, overlap directly with one of our ILT dimensions and others, such as Humane-oriented, likely overlap with Sensitivity. In addition, recent work by Abdulhadi and Schyns (2015) in Saudi Arabia found that their reported

items reflected all of Offermann et al.'s (1994) dimensions except masculinity. Nonetheless, examining ILTs within specific cultural contexts is important in understanding local views of leaders, and would be an important area for future research.

Future research should also seek to further investigate gender and racial differences in leader perceptions. Our post hoc analyses showed that women raters tended to see leaders somewhat more positively on a number of dimensions, though gender differences were small in magnitude. Nonetheless, this is compatible with the view that although men and women may not differ in evaluating their actual supervisors, women may rate others more leniently than men in situations where performance data are lacking (Shore & Thornton, 1986). They also rated masculinity significantly lower as a prototypical characteristic of leaders than male raters did. In addition, post-hoc factor analyses suggests that men and women may differ in their views of charisma, with women seeing it as more dynamic and strength-related, and men seeing it as more of an effective public communicator persona that is different from strength and command. Given the small number of racial minorities in our sample, we did not conduct cross-racial comparisons of leader ratings, but would encourage future researchers with access to larger minority populations to do so.

Conclusion

Overall, the emergence of ILT factors in 2014 closely mirrors that found 20 years ago, lending support to the idea that these factors of implicit theories of leadership are remarkably stable despite many organizational and societal changes. The primary change in ILT content was one of increased emphasis rather than addition or replacement, whereby Creativity emerged as a factor in its own right in the present samples where it had not done so in earlier analyses. The increasing prominence of creativity in the recent samples may be a result of U.S. economic forces emphasizing innovation as the cure for previous economic malaise, and the role of leaders in developing climates that encourage and promote discovery and invention. This view of creativity as a key leadership factor was particularly reported by men. Although naïve ILT associations of women with leadership still have not surfaced, there are some indications that the gendered-ness of leadership may be dissipating somewhat in that key leadership dimensions such as dedication and intelligence fail to correlate with any gender items, and that others that correlate with masculine items also correlate with feminine items, albeit at a lower level. Perhaps one marker of women's full public acceptance into organizational leadership roles will be when individuals include them into their naïve conceptions of leadership, or perhaps do not include reference to either gender in their ILTs.

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